

Day_4/hw2 증거자료

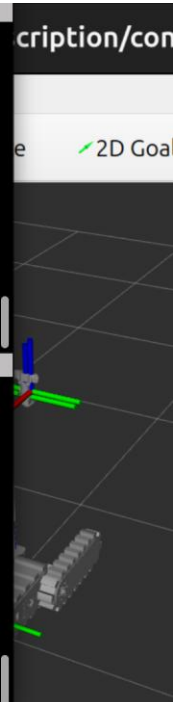
김선우

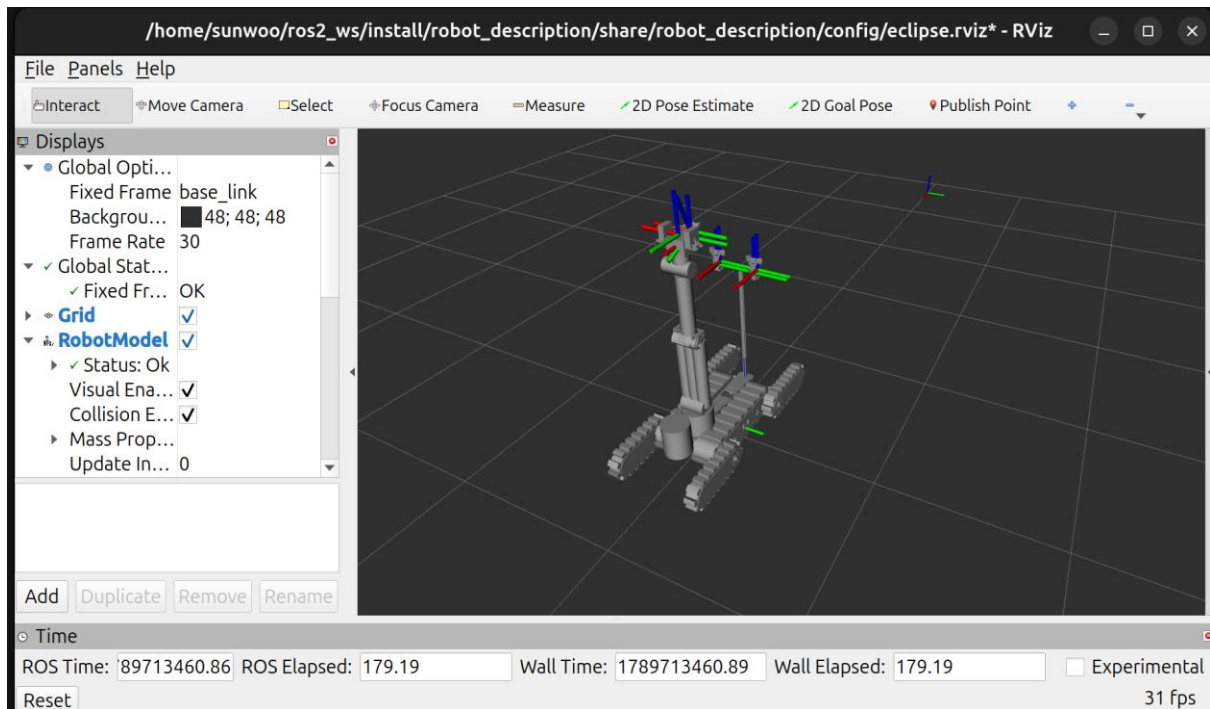
ros2 control list_controllers 실행 결과, joint_state_broadcaster, arm_controller, camera_tower_controller, base_drive_controller, flipper_controller 5개 컨트롤러가 모두 active 상태임을 확인하였다. 이후 ros2 control list_hardware_components 결과에서도 eclipse_base, eclipse_arm, eclipse_arm2, eclipse_camera_tower 4개 하드웨어 컴포넌트(하드웨어 하나하나 ex.몸, 타워, 바퀴 등)가 모두 mock_components/GenericSystem 플러그인(하드웨어를 어떻게 제어할지에 대한 코드?)으로 active 상태임을 확인

```
sunwoo@sunwoo-940XHA: ~/ros2_ws 39x11 | sunwoo@sunwoo-940XHA: ~/ros2_ws 39x11
velocity command timed out. Braking.
ros2_control_node-2] [WARN] [178971348
.591621342] [base_drive_controller]: V
velocity command timed out. Braking.
ros2_control_node-2] [WARN] [178971348
.601621078] [base_drive_controller]: V
velocity command timed out. Braking.
ros2_control_node-2] [WARN] [178971349
.611526961] [base_drive_controller]: V
velocity command timed out. Braking.
sunwoo@sunwoo-940XHA:~/ros2_ws$

sunwoo@sunwoo-940XHA:~/ros2_ws 39x11 | sunwoo@sunwoo-940XHA: ~/ros2_ws 39x11
sunwoo@sunwoo-940XHA:~/ros2_ws$ source
/rosws/install/setup.bash
ros2 topic echo /joint_states --once
message was lost!!!
total count change:1
total count: 1--
header:
stamp:
sec: 1789713325
nanosec: 691532003
frame_id: base_link
publisher: beginning loop
publishing #1: geometry_msgs.msg.TwistS
tamped(header=std_msgs.msg.Header(stamp
=builtin_interfaces.msg.Time(sec=0, nan
osec=0), frame_id='base_link'), twist=g
eometry_msgs.msg.Twist(linear=geometry_
msgs.msg.Vector3(x=0.5, y=0.0, z=0.0),
angular=geometry_msgs.msg.Vector3(x=0.0
, y=0.0, z=0.0)))
sunwoo@sunwoo-940XHA:~/ros2_ws$

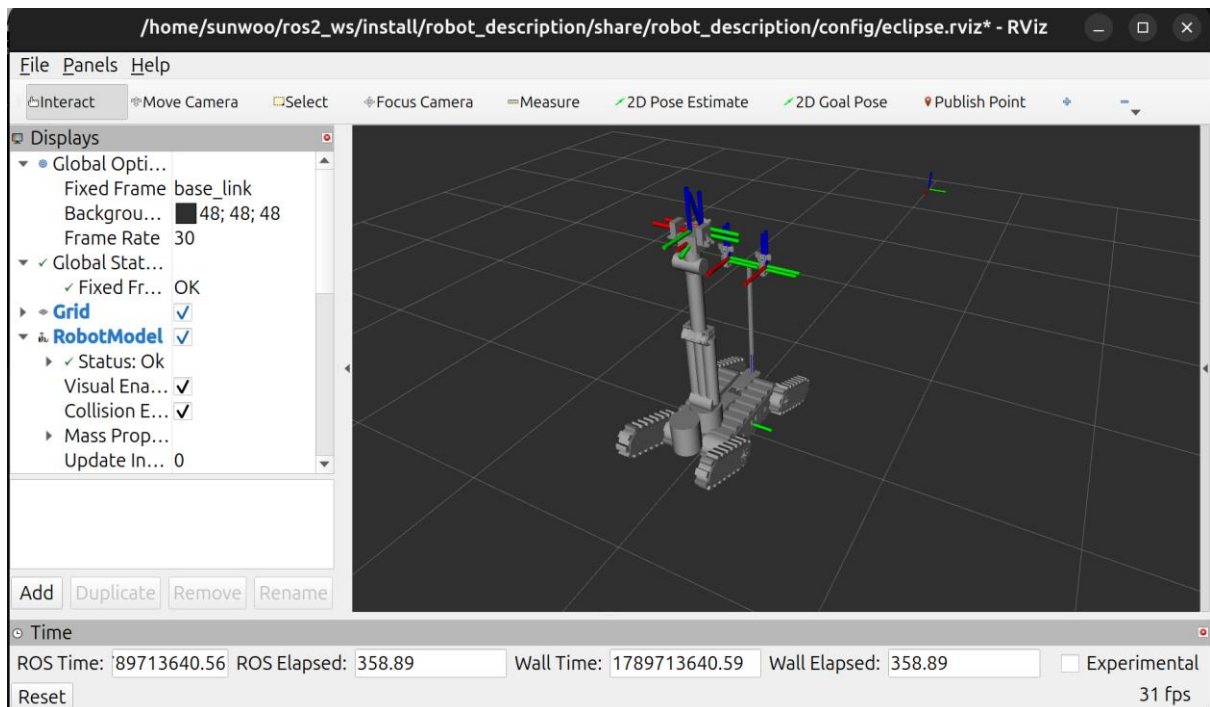
sunwoo@sunwoo-940XHA:~/ros2_ws 39x11 | sunwoo@sunwoo-940XHA: ~/ros2_ws 39x11
publishing #84: geometry_msgs.msg.Twist
Stamped(header=std_msgs.msg.Header(stam
p=builtin_interfaces.msg.Time(sec=0, na
nosec=0), frame_id='base_link'), twist=
geometry_msgs.msg.Twist(linear=geometry
_msgs.msg.Vector3(x=0.5, y=0.0, z=0.0),
angular=geometry_msgs.msg.Vector3(x=0.
0, y=0.0, z=0.0)))
^Csunwoo@sunwoo-940XHA:~/ros2_ws$
```



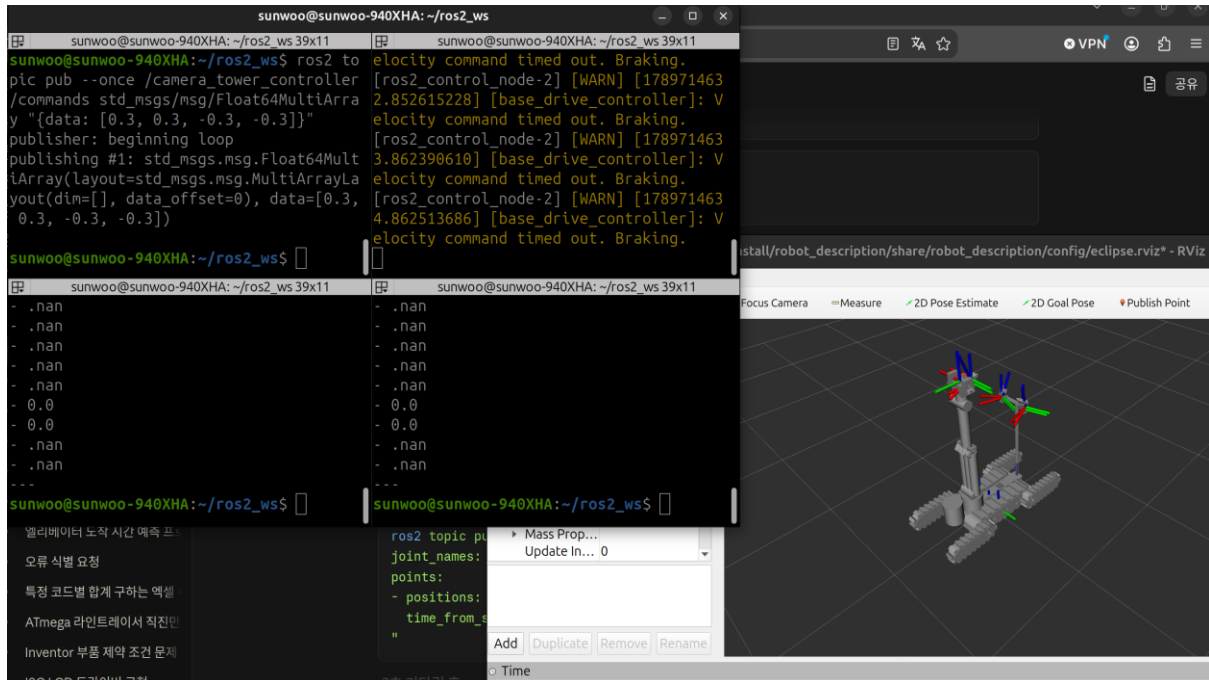


바퀴 움직임.

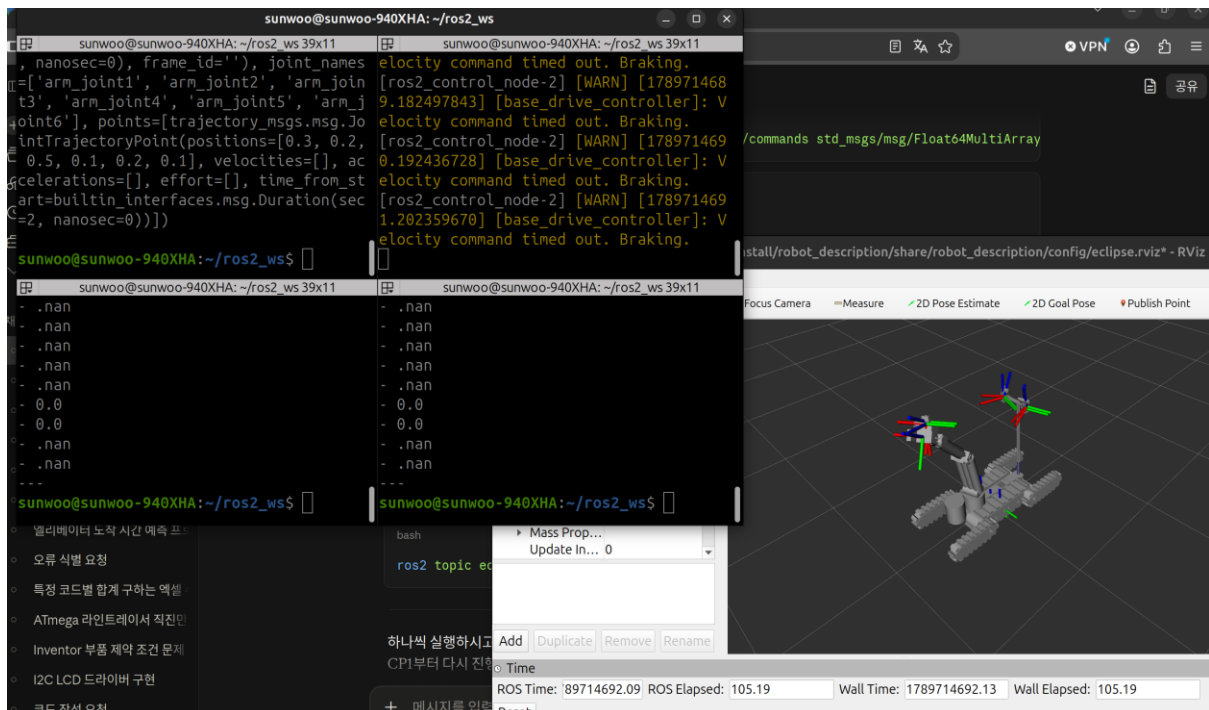
바퀴 축과 로봇이 원래 위치해있던 원점의 위치를 보면 로봇이 이동했음을 알 수 있다.



플리퍼 움직임



카메라타워 회전(joint_states 확인 결과, ct_joint1_1=0.3, ct_joint2_1=0.3, ct_joint1_2=-0.3, ct_joint2_2=-0.3으로 명령값이 정확히 반영됨을 확인)



팔 움직임